

Scalability & Future Plan

Date	8 July 2024
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local Database Architecture (SQLite)	SQLite locks the entire database during writes. It cannot handle thousands of concurrent students logging emotions simultaneously.	High
2	External API Rate Limits (Google Gemini)	If the user base spikes rapidly, we may hit the free-tier request limits of the Gemini API, causing coaching responses to fail.	Medium
3	Local Model Inference	Running the BiLSTM/BERT models locally consumes high RAM/CPU on the host machine, limiting hardware deployment options.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Running locally on a single Streamlit server thread.	Dockerize the application and deploy it to a load-balanced cloud provider like AWS Elastic Beanstalk or Heroku.
2	Data Storage	Local SQLite .db file.	Migrate from SQLite to a highly available, cloud-hosted PostgreSQL or MongoDB database.
3	Performance	Deep learning models run on standard CPU architecture.	Move ML inference to AWS SageMaker or deploy the models on GPU-backed cloud instances for instantaneous prediction.
4	Security	Basic bcrypt password hashing with local .env key storage.	Implement OAuth 2.0 (Google/Microsoft Login) and migrate API keys to a secure vault like AWS Secrets Manager .

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment & DB Migration	1-2 Months	Transitions the prototype into a globally accessible, production-ready web application.
Phase 2	Multi-Language Support	3-4 Months	Translates the UI and utilizes Gemini's multilingual capabilities to support non-English speaking students worldwide.
Phase 3	Voice-to-Text Emotion Detection	6 Months	Allows students to speak instead of type. Voice tone analysis will drastically improve the accuracy of emotion detection.
Phase 4	LMS Integration (Canvas/Moodle)	9-12 Months	Transforms the app into a B2B SaaS product that universities can plug directly into their existing course portals.